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PYTHON FUNDAMENTALS

Lecture 2 — Python Core, Ecosystem & Tools

Top 3 Languages • Why Python? • Syntax
Applications • Global Tech Giants
IDEs (PyCharm, VS Code, Anaconda, NumPy)
Google Colab • Open Source Superpowers

COMPLETE HANDS-ON STUDY NOTES

01

Top 3 Programming Languages

Python, JavaScript and C/C++ dominate global software.

- **Python** — the AI, data and automation standard
- **JavaScript** — the universal language of the web
- **C / C++** — the high-performance system engine
- Python ranks #1 on TIOBE and GitHub indices
- JavaScript powers 98% of client-side web apps

01

Python — Purpose & Examples

The easiest of the three to learn.

- **Purpose** — AI, ML, data analytics, backend APIs
- **Example** — ChatGPT algorithms
- **Example** — YouTube recommendation engine
- **Strength** — simple, English-like grammar
- **Learning curve** — easiest, beginner friendly

01 JavaScript — Purpose & Examples

The mandatory language for front-end engineers.

- **Purpose** — interactive websites and mobile apps
- Also runs servers through Node.js
- **Example** — Google Maps interactivity
- **Example** — Netflix browser interface
- **Learning curve** — moderate (asynchronous concepts)

01 C / C++ — Purpose & Examples

Direct hardware access and memory control.

- **Purpose** — operating systems, embedded chips, games
- Runs kernels, game engines, flight software
- **Example** — Windows OS kernel
- **Example** — Unreal Engine 5 games
- **Learning curve** — steep (pointers, manual memory)

01 Why We Start With Python

Key takeaway for students.

- Python removes painful syntax hurdles
- Master algorithmic thinking first
- Low-level memory complexity comes later
- The same basics lead to many careers

02 Why Python? Core Significance

Built for productivity and readability.

- **Creator** — Guido van Rossum, 1991
- **Principle** — code is read more than it is written
- **Clean indentation** — no semicolons or curly braces
- **Interpreted** — runs line by line, no recompiling
- **Portable** — Windows, macOS, Linux, Raspberry Pi

02 Dynamic Typing & Productivity

Write less code, get more done.

- **Dynamic typing** — no need to declare variable types
- **Example** — C++ needs int x = 10; Python: x = 10
- Python detects the type automatically
- **Fewer lines** — 15–20 lines in Java/C++ become 2–3
- **Result** — faster development and cleaner code

02 What Is Python? Definition

A one-line summary for your notes.

- **Interpreted** — runs without a compile step
- **Object-oriented** — programs are built from objects
- **High-level** — human-friendly, hides hardware details
- Dynamic typing and dynamic binding
- Ideal for Rapid Application Development

02 Example: Swapping Variables

Python vs C++/Java — one clean line.

```
# Traditional C++/Java needs a temporary variable:
# temp = a; a = b; b = temp;

# In Python, it takes just one line:
a = 10
b = 20
a, b = b, a   # a is now 20, b is now 10
print(f"Value of a: {a}, Value of b: {b}")
```

REMEMBER

Python swaps two variables in a single line.

03 Major Applications of Python

From web servers to autonomous robotics.

- Artificial Intelligence & Machine Learning
- Data Science & Big Data Analysis
- Full-Stack Web & API Development
- Automation & Web Scraping
- Cybersecurity & Ethical Hacking

03 AI, ML & Data Science

Applications with real examples.

- **AI & ML** — train GenAI, NLP and vision models
- **Libraries** — PyTorch and TensorFlow
- **Data Science** — clean messy real-world datasets
- **Libraries** — Pandas, NumPy, Matplotlib
- **Example** — compute statistics, draw trend charts

03 Web, Automation & Security

Applications with real examples.

- **Web** — Django (secure, batteries-included)
- **APIs** — FastAPI (ultra-fast async APIs)
- **Automation** — rename files, send bulk emails
- **Scraping** — prices via Beautiful Soup, Selenium
- **Security** — network scanners, sniffers (Scapy)

03 Example: Fetching Web Data

A practical 3-line program.

```python
# Practical 3-line web data fetching
import requests
response = requests.get("https://api.github.com")
print("GitHub Server Status:", response.status_code)
# Outputs: 200 (Success)
```

WHY IT MATTERS

One language opens many careers — Data, ML, Web, DevOps, Security.

04

Tech Giants Using Python (1/2)

Market leaders trust Python at scale.

- **Google** — search, YouTube, Cloud, DeepMind, Gemini
- **Google** — an official server-side language
- **Netflix** — personal recommendation algorithms
- **Netflix** — video encoding and content delivery
- **Instagram** — world's largest Django deployment

04 Tech Giants Using Python (2/2)

More companies, more examples.

- **Spotify** — audio streaming backend and analytics
- **Spotify** — playlist curation, radio recommendations
- **OpenAI** — ChatGPT, GPT-4o, Whisper, DALL-E
- **OpenAI** — built and deployed with Python and PyTorch
- **NASA & CERN** — James Webb, Large Hadron Collider

04

Python at Scale

Impact and engineering scale.

- **Google** — billions of real-time search queries daily
- **Instagram** — 2B+ active users sharing photos/reels
- **Netflix** — streaming for 260M+ paid accounts
- **OpenAI** — global standard for Generative AI
- Python scales to billions of users

05 Python IDEs & Tools

Editors, IDEs, data suites and libraries.

- **PyCharm (JetBrains)** — full-fledged professional IDE
- **VS Code (Microsoft)** — lightweight, modular editor
- **Anaconda** — Python plus data science libraries
- **NumPy** — a library, not an editor or IDE
- Choose the tool that fits your project

05

PyCharm vs VS Code

Choose by project size and style.

- **PyCharm** — smart code completion, visual debugging
- **PyCharm** — refactoring and database tools
- **Best for** — large commercial Python projects
- **VS Code** — Jupyter notebooks, Git, extensions
- **Best for** — daily coding, web and scripts

05

Anaconda & NumPy

Data science essentials.

- **Anaconda** — complete data science platform
- Pre-packaged with 250+ data libraries
- **conda** — package manager avoids conflicts
- **NumPy** — fast multi-dimensional arrays, matrices
- **NumPy** — up to 50x faster than standard lists

05 Example: Why NumPy?

NumPy beats standard Python lists.

```python
import numpy as np
numbers = np.array([10, 20, 30, 40, 50])

# Vectorized: multiplies all items at C-speed, no loop!
result = numbers * 2
print("Doubled Array:", result)
# Output: [20, 40, 60, 80, 100]
```

KEY IDEA

Vectorized math works on whole arrays — no loops needed.

06

What Is Google Colab?

Run Python in the cloud — no installation.

- Hosted Jupyter notebook service by Google
- Runs entirely in your web browser
- **Zero setup** — start coding in 10 seconds
- Share a notebook with a single URL
- Works just like Google Docs

06 Colab Superpowers

Free tools for beginners and engineers.

- **Free GPU and TPU** — NVIDIA T4, Google TPUs
- Train AI models without costly hardware
- **One document** — code, text, LaTeX and charts
- **Google Drive** — files auto-save securely
- **Install packages** — !pip install <library>

06 Open Source Superpowers

Freedom to learn, build and share.

- Python and its ecosystem are fully open source
- Inspect and modify the source code freely
- Build commercial products with $0 license fees
- **PyPI** — 500,000+ ready-made packages
- **Example** — face detection, PDFs, stock tracking

07

Lecture 2 — Key Takeaways

Quick revision.

- **Top 3** — Python, JavaScript, C/C++
- **Python** — simple, readable, productive
- **Uses** — AI, data, web, automation, security
- **Tools** — VS Code, PyCharm, Anaconda, NumPy
- **Colab and open source** — learn at zero cost

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THANK YOU

Keep learning. Keep practicing.

Keep building with Python!



See you in Lecture 3 —
Variables, Data Types & Operators